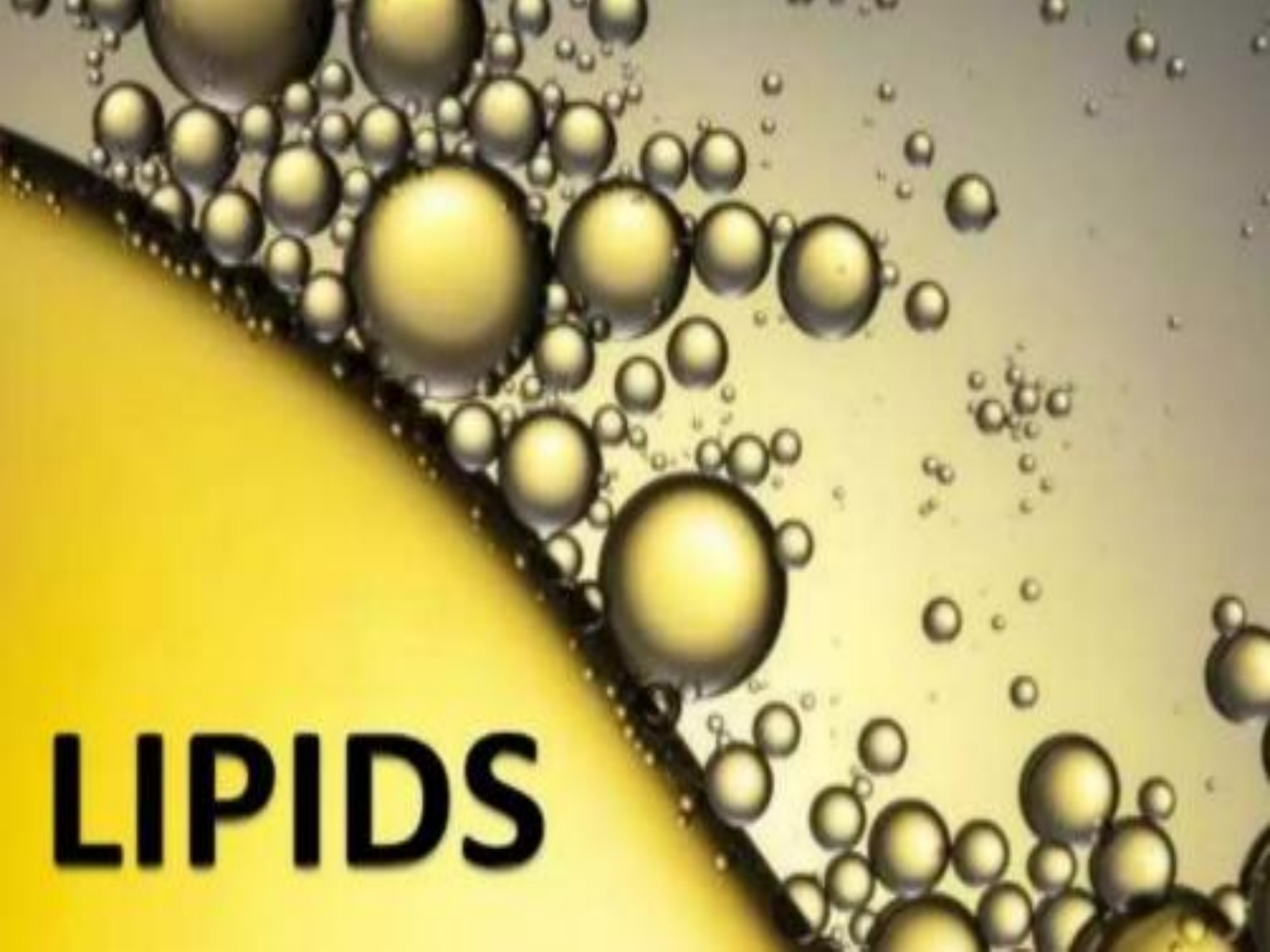
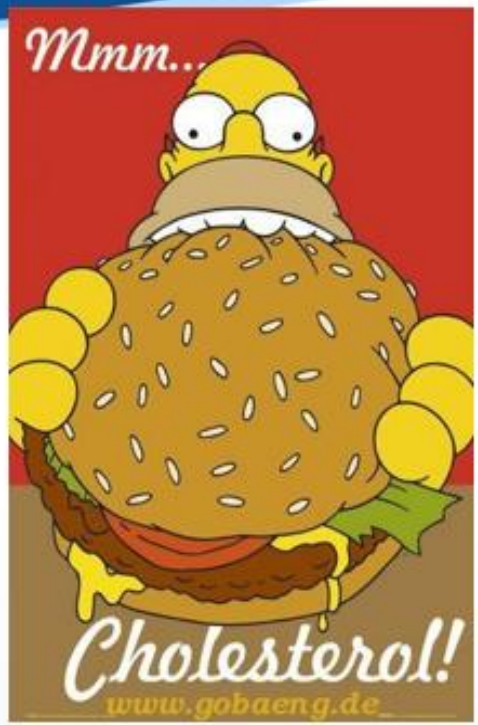

LIPIDS

LIPIDS

- Lipids are a diverse group of organic compounds
- Lipids contain carbon,hydrogen,oxygen,nitrogen and phosphorous
- Lipids are non-polar and hydrophobic compounds i.e insoluble in water
- They are easily soluble in organic solvents like ether, alcohol, chloroform, benzene etc
- Lipids store double the amount of energy as compared to carbohydrates and proteins because
- They contain high proportion of C-H bonds & very low oxygen as compared to carbohydrates and proteins
- Act as insulating layer e.g. waxes in exoskeleton of insects

Lipids



- **Function**: Store energy, Insulates your body, and make up the cell membrane!
- **Elements**: C-H-O
- **Monomer (Building blocks)**: glycerol & 3 fatty acids
- **Polymer**: Phospholipids, triglycerides
- **Example**: Steroids, cholesterol, fats, Oils, Nuts, Waxes, and make up part of the cell membrane!

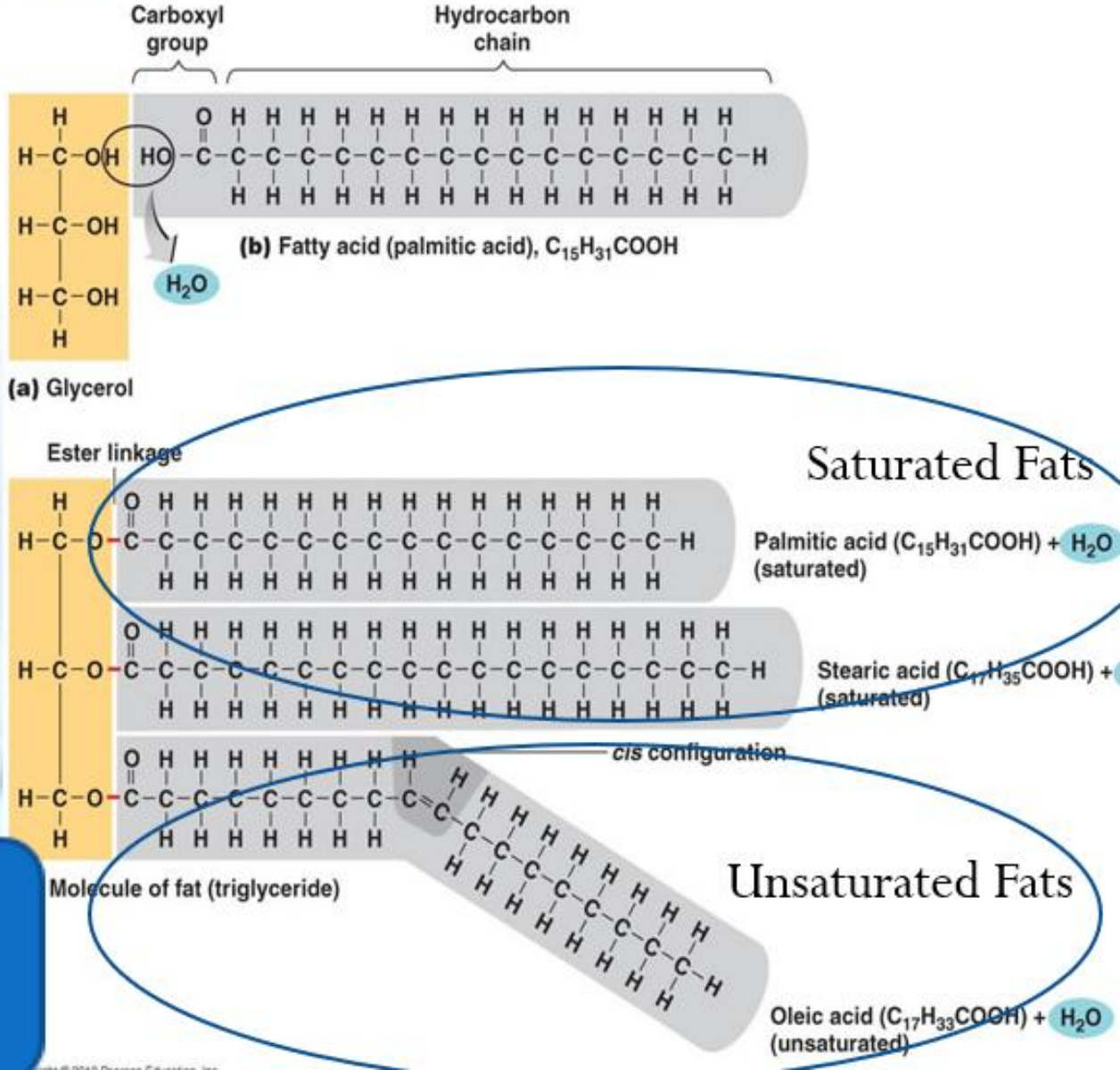


Biomedical Importance

- Major source of energy for the body (High caloric value= 9.3 cal/g)
- Important dietary constituent, fat soluble vitamins & essential fatty acid are contained in the fat of natural food
- Fat is stored in adipose tissue
- Serve as thermal insulator in subcutaneous tissues & around certain organs
- Act as electrical insulator, allowing rapid propagation of depolarization waves along myelinated nerves
- Combination of lipid & protein (LP) serve as the means of transporting lipid in blood

Lipid Structure

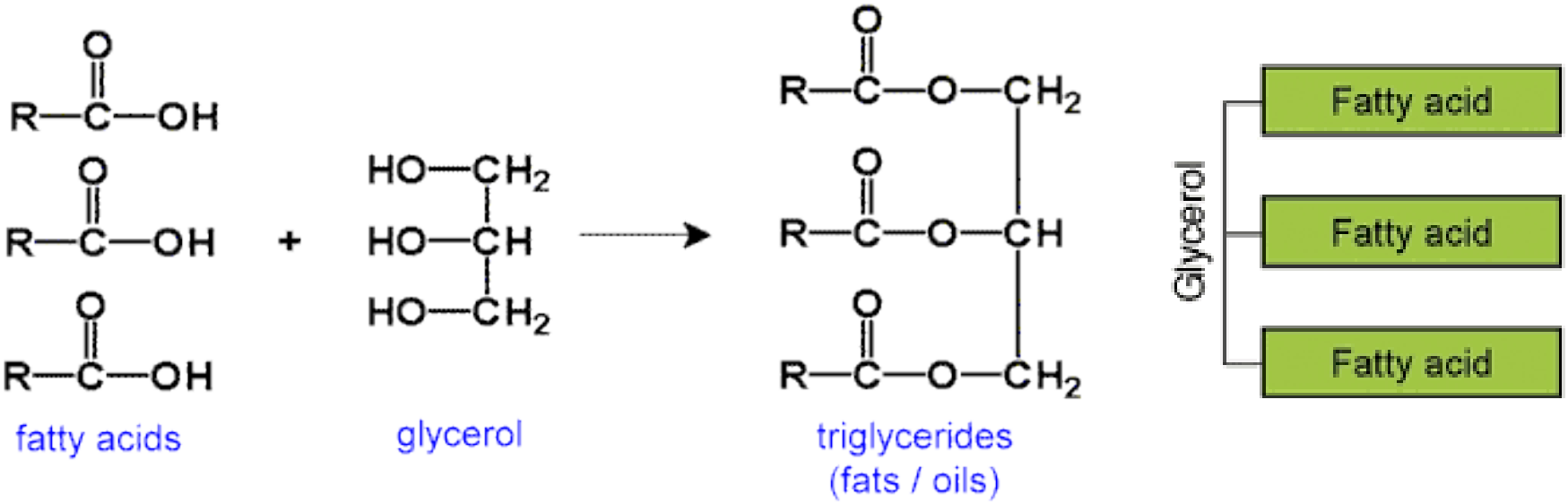
- Remember: Elements present are C, H, O
- Long strands of Carbon and Hydrogen



CALLED HYDROCARBONS!

LIPIDS

- Lipids are esters of **fatty acids with alcohol**



“R” group is made up of hydrocarbon chains

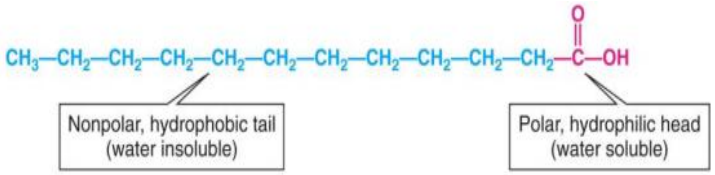
What are fatty acids and its types?

Fatty Acids

carboxylic acid group (COOH)
joined to a long tail of
carbon and hydrogen atoms

The **length** of the **hydrocarbon tail** varies, giving rise to the various fatty acids.

The tail is normally written as **R**, giving the formula **R.COOH**





- **Properties**
  - Saturated fatty acids are solid at room temperature and have a high melting point
  - Unsaturated fatty acids are liquid at room temperature and have a low melting point

Saturated fatty acid (no double bonds)



- Butyric acid: CH3-CH2-CH2-COOH
- Caprylic acid: CH3(CH2)8 COOH
- Palmitic acid:CH3(CH2)14 COOH
- Stearic acid: CH3(CH2)16 COOH

Unsaturated fatty acid

1. Monounsaturated fatty acid (single double bond)



a.) oleic acid: C18 H34 O2
b.) palmitolic acid: C16 H30 O2

2. Polyunsaturated fatty acid (more than one double bond)



a.) linolic acid: C18 H32 O2
b.) eleostearic acid: C18 H30 O2
c.) arachidonic acid: C20 H32 O2

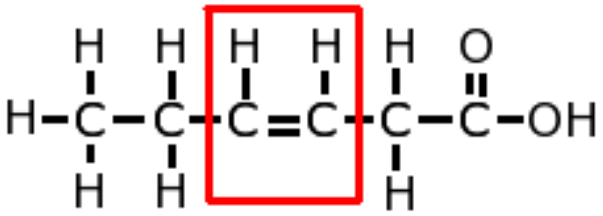
PUFA & MUFA

FATTY ACIDS STRUCTURE

Unsaturated fatty acids

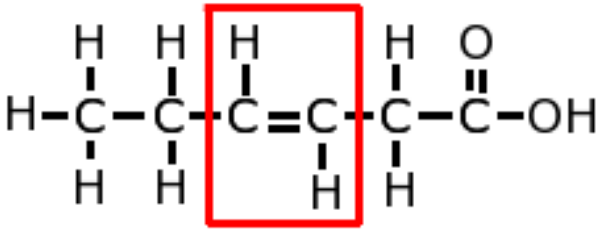
Cis: hydorgens of the carbons with the double bond are **on the same side** of their respective carbons

Trans: Hydrogens of the carbons with the double bond are **on opposite sides** of their respective carbons



Cis Fatty Acid

CIS

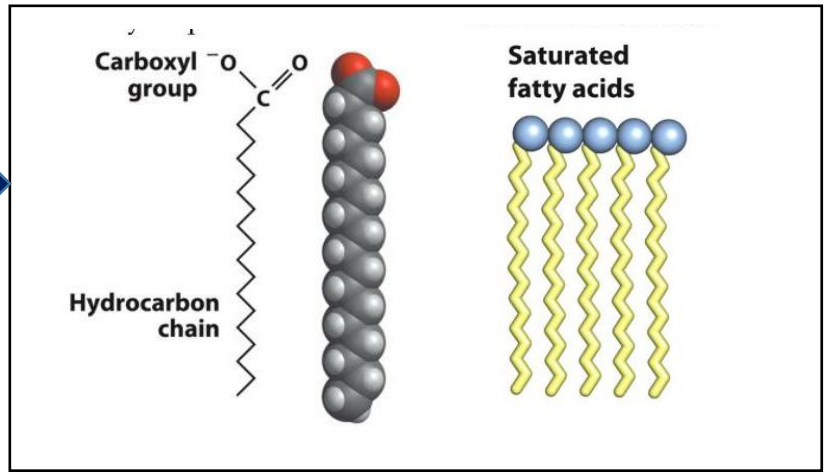


Trans Fatty Acid

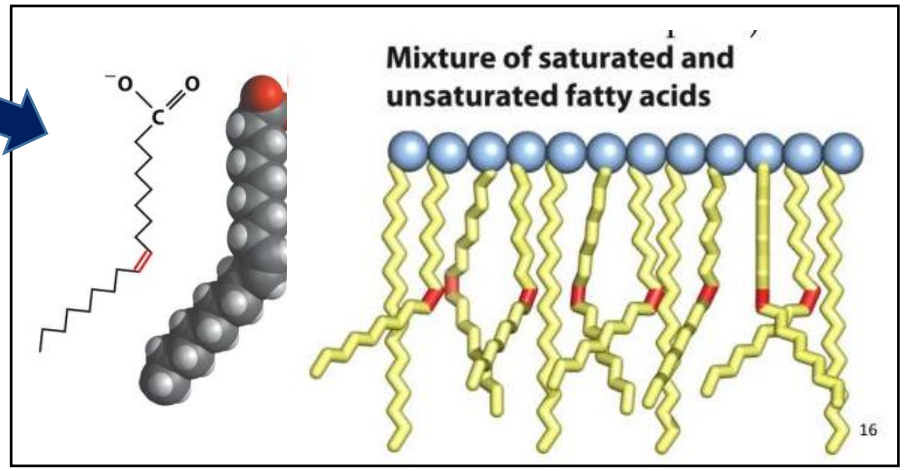
TRANS

Why saturated fatty acids are solid and have high melting point than unsaturated fatty acids?

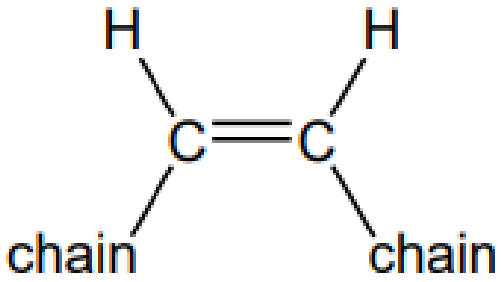
- Hydrocarbon chains of saturated fatty acids can lie parallel with strong dispersion forces between their chains; they pack into well-ordered, compact crystalline forms and melt above room temperature.



- Because of the *cis* configuration of the double bonds in unsaturated fatty acids, their hydrocarbon chains have a less ordered structure and dispersion forces between them are weaker; these triglycerides have melting points below room temperature.



- Requires less thermal energy to disrupt (lower melting point)



Physical properties of fatty acids

The physical properties of the fatty acids, and of compounds that contain them, are largely determined by the length and degree of unsaturation of the hydrocarbon chain.

A. The nonpolar hydrocarbon chain accounts for the poor solubility of fatty acids in water. Solubility decreases:
   - With longer fatty acyl chain
   - With fewer double bonds

B. Melting points are also strongly influenced by:
   - The length of the hydrocarbon chain
     ↑ length, ↑ melting point
   - Degree of unsaturation
     ↑ unsaturation, ↓ melting point

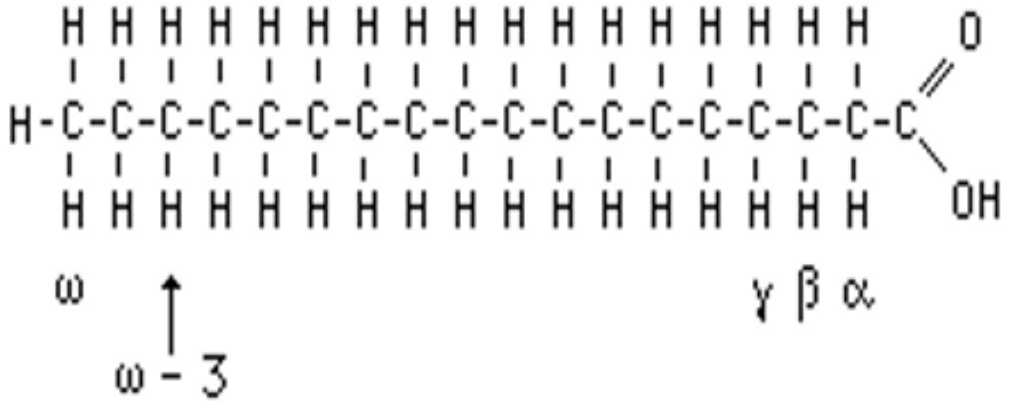
Some chemistry of common fatty acids

#C's	Name	Formula	MP	Common Sources
Saturated				
14	Myristic acid	$CH_3(CH_2)_{12}COOH$	54°C	Butterfat, coconut oil, nutmeg oil
16	Palmitic acid	$CH_3(CH_2)_{14}COOH$	63°C	Lard, beef fat, butterfat, cottonseed oil
18	Stearic acid	$CH_3(CH_2)_{16}COOH$	70°C	Lard, beef fat, butterfat, cottonseed oil
20	Arachidic acid	$CH_3(CH_2)_{18}COOH$	76°C	Peanut oil
Monounsaturated				
16	Palmitoleic acid	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$	-1°C	Cod liver oil, butterfat
18	Oleic acid	$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	13°C	Lard, beef fat, olive oil, peanut oil
Polyunsaturated				
18	Linoleic acid	$CH_3(CH_2)_4(CH{=}CHCH_2)_2(CH_2)_6COOH$	-5°C	Cottonseed oil, soybean oil, corn oil, linseed oil
18	Linolenic acid	$CH_3CH_2(CH{=}CHCH_2)_3(CH_2)_6COOH$	-11°C	Linseed oil, corn oil
20	Arachidonic acid	$CH_3(CH_2)_4(CH{=}CHCH_2)_4(CH_2)_2COOH$	-50°C	Corn oil, linseed oil, animal tissues

What patterns do you observe?

FATTY ACIDS - NOMENCLATURE

Greek Letters are Used in Two Ways



Use of Greek letters to designate carbons

The carbon next to the -COOH group is designated α; the next one is β, and so forth. The most distant carbon is designated ω. Sometimes carbon atoms close to the ω carbon are designated in relation to it. *E.g.*, the third from the end is ω - 3 (omega minus 3).

FATTY ACIDS - NOMENCLATURE

Summary of the Four Common Systems

Here are the four commonly used ways of designating fatty acids.

- Linoleic(18:2 Δ 9,12)
- Linolenic (18:3 Δ 9,12,15)
- Arachidonic acid (20:4 Δ 5,8,11,14)

Names		Abbreviations	
trivial	IUPAC	carboxyl-reference	ω - reference
palmitic acid	hexadecanoic acid	16:0	16:0
stearic acid	octadecanoic acid	18:0	18:0
oleic acid	9-octadecenoic acid	18:1 Δ^{9}	18:1 (ω-9)
linoleic acid	9,12-octadecenoic acid	18:2 $\Delta^{9,12}$	18:2 (ω-6)
linolenic acid	9, 12, 15-octadecenoic acid	18:3 $\Delta^{9,12,15}$	18:3 (ω-3)

Lipids classification

Simple lipids

They are esters of fatty acids with glycerol.

1. **True fats and oil** (alcohol is glycerol)- also called as glycerides
2. **Waxes** (alcohol is other than glycerol. Example is beeswax)

Conjugated or compound lipids or complex lipids

They are esters of fatty acids with glycerol or other alcohol plus other groups

1. Glycerophospholipids
2. Sphingophospholipids
3. Sphingoglycolipids
4. Lipoproteins

Derived lipids

They are obtained on hydrolysis of simple and complex lipids

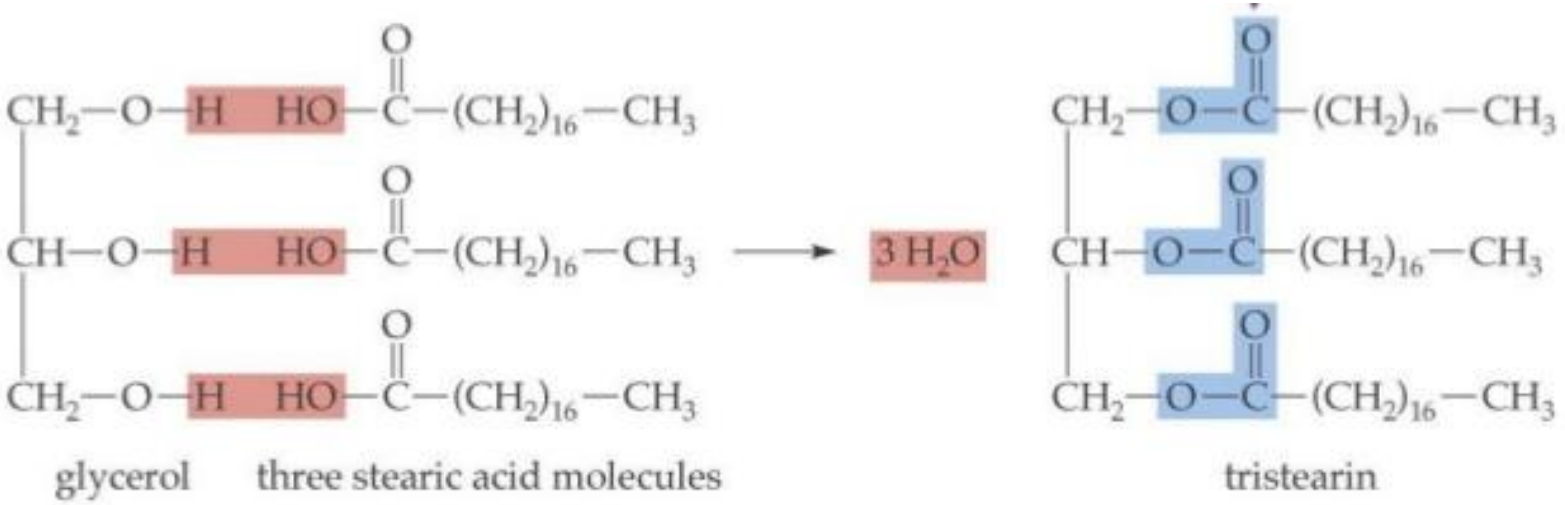
1. Cholesterol
2. Retinol
3. Steroids
4. Hormones
5. Fat soluble vitamins
6. Ketone bodies

Simple Lipids

- Animal fats and vegetable oils are esters composed of three molecules of a fatty acid connected to a glycerol molecule, producing a structure called a **triglyceride** or a **triacylglycerol**:

Fats – also known as triglyceride or triacylglycerol

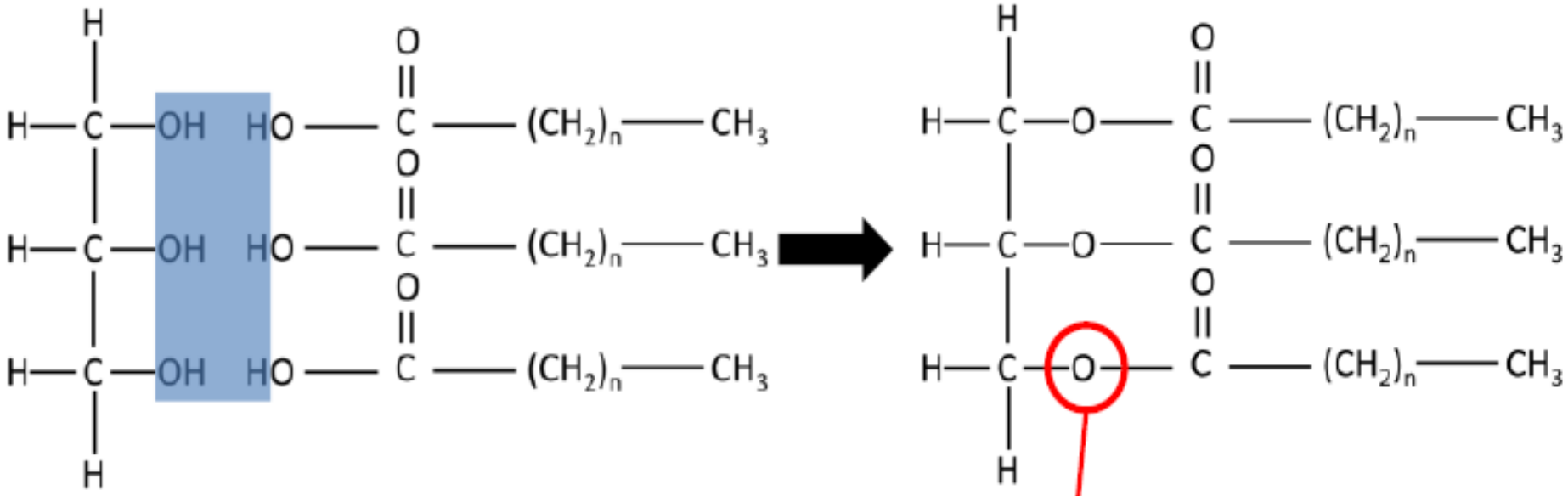
Example: Tristearin (3 molecules of stearic acid + one molecule of glycerol).



Tristearin

Triglycerides formation

Condensation reaction between glycerol and fatty acids



Glycerol Three Fatty Acids Triglyceride

Lipids are glycerol combined with 1, 2 or 3 fatty acids, therefore **triglycerides are lipids**

n.b. hydrolysis *is the reverse of this process, catalysed by lipase*

+ $3H_2O$

Covalent bonds called ester bonds are formed between the fatty acids and glycerol molecules.

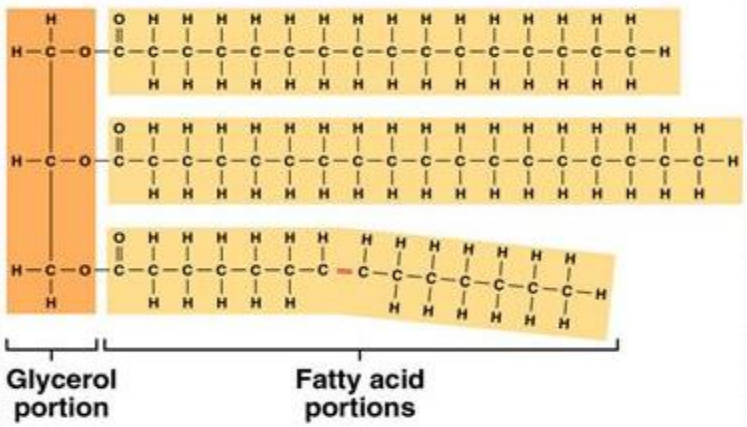


- *Oils*: Triglycerides rich in unsaturated fatty acids are generally liquid at room.
- *Fats*: Triglycerides rich in saturated fatty acids are generally semisolids or solids at room temperature.

Fats & Oils (triglycerides)- long term energy storage

•Fat has twice the calories of carbohydrates.

– fat = 9 cal/g sugar= 4 cal/g



Health tip:

Saturated or hydrogenated fats(bad) vs. unsaturated (good)



Saturated Fat

meats, butter,
dairy products

solid at room
temperature

*** increase levels of
"bad" cholesterol
(low-density lipoprotein)

low-density lipoprotein
clogs arteries



Unsaturated Fat

vegetable oils

liquid at room
temperature

increase levels of ***
"good" cholesterol
(high-density lipoprotein)

high-density lipoprotein,
or HDL, "grabs" LDL
and escorts it to the liver
where **LDL is broken down**
and eventually removed

Conjugated lipids

- Conjugated or compound lipids or complex lipids
- Esters of fatty acids with either glycerol and/or other alcohol plus other groups

1. Glycerophospholipids
2. Sphingophospholipids

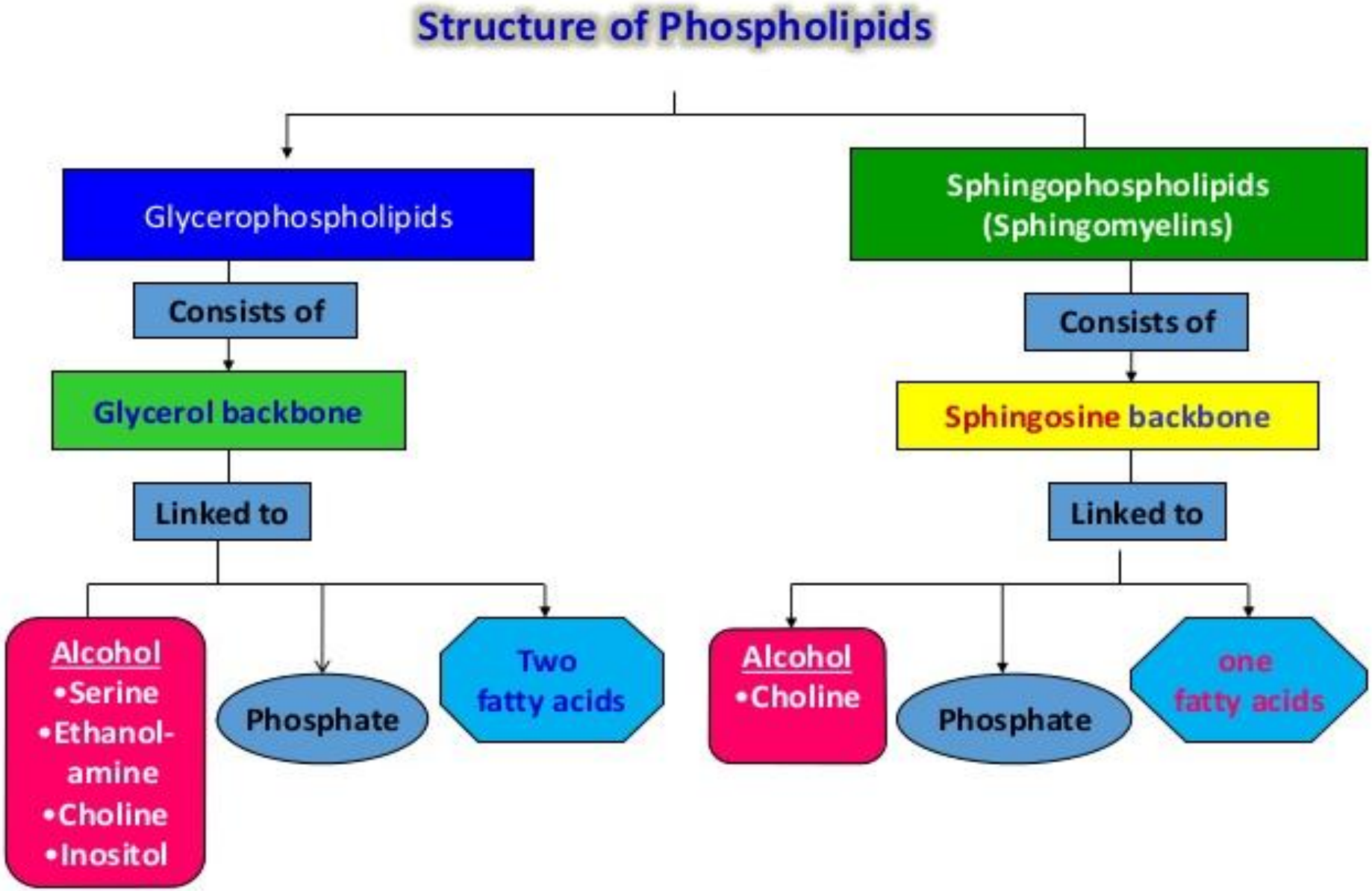
} Phospholipids

3. Sphingoglycolipids
4. Lipoproteins

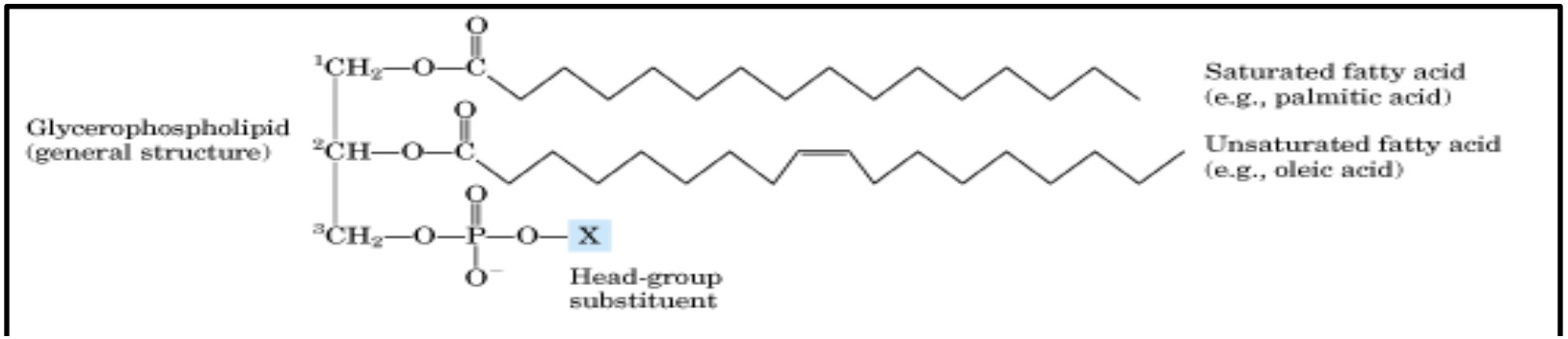
Phospholipids

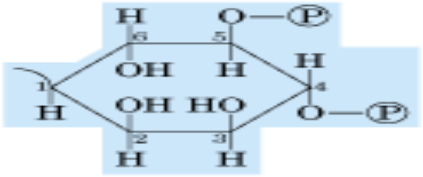
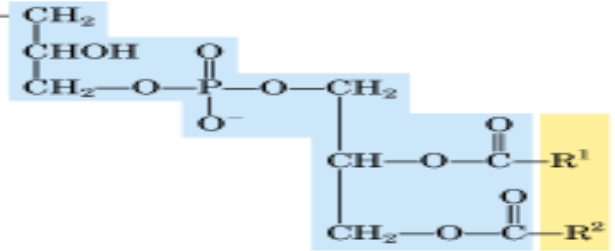
- A class of lipids that are a major component of all cell membranes as they can form lipid bilayers.
- Composition: phospholipids are composed of
- i. Glycerol: one molecule
- ii. Fatty acids :Two molecules
- iii. Phosphoric acid: one molecule
- When a nitrogen containing group is attached with phosphate end of phospholipids, it is known as Phosphatidyl Choline.
- Phospholipids have two parts
- i) Head:
- Head is polar in nature, Soluble in water (Hydrophilic).
- ii) Tails:
- Tail is non-polar in nature, insoluble in water (Hydrophobic)

Phospholipids



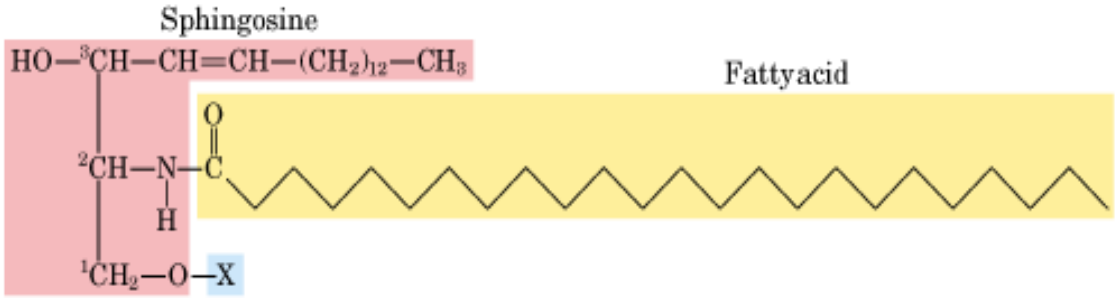
Structure of glycerophospholipid

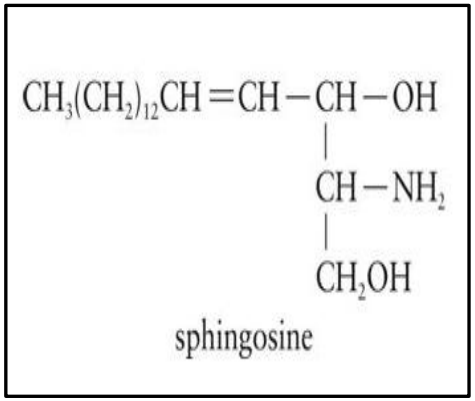


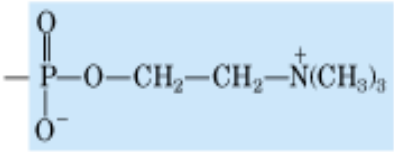
Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	—H	−1
Phosphatidylethanolamine	Ethanolamine	$-CH_2-CH_2-\overset{+}{N}H_3$	0
Phosphatidylcholine	Choline	$-CH_2-CH_2-\overset{+}{N}(CH_3)_3$	0
Phosphatidylserine	Serine	$-CH_2-CH(COO^-)-\overset{+}{N}H_3$	−1
Phosphatidylglycerol	Glycerol	$-CH_2-CH(OH)-CH_2-OH$	−1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		−4
Cardiolipin	Phosphatidyl-glycerol		−2

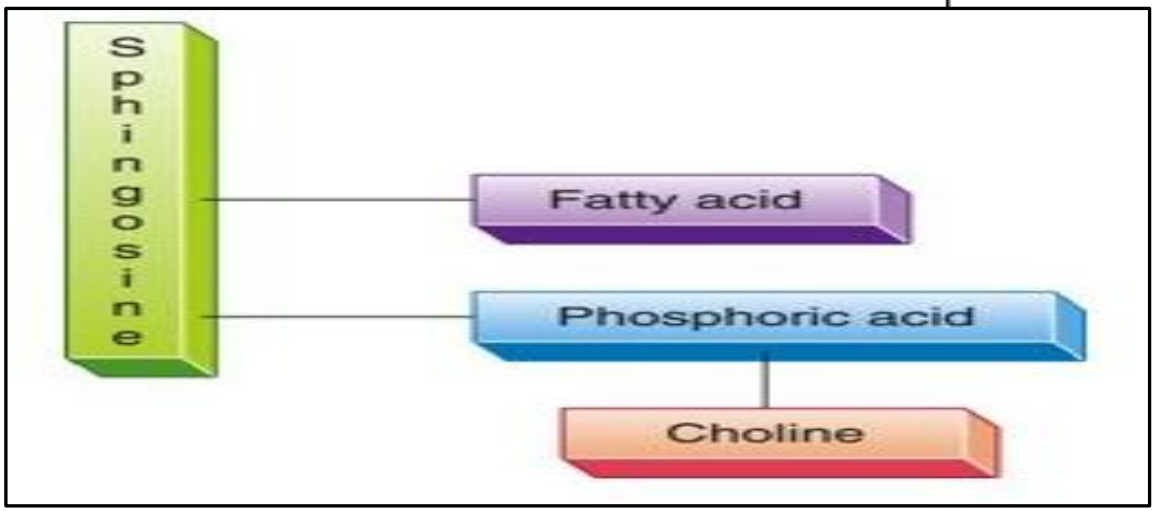
Structure of sphingophospholipid

Sphingolipid (general structure)





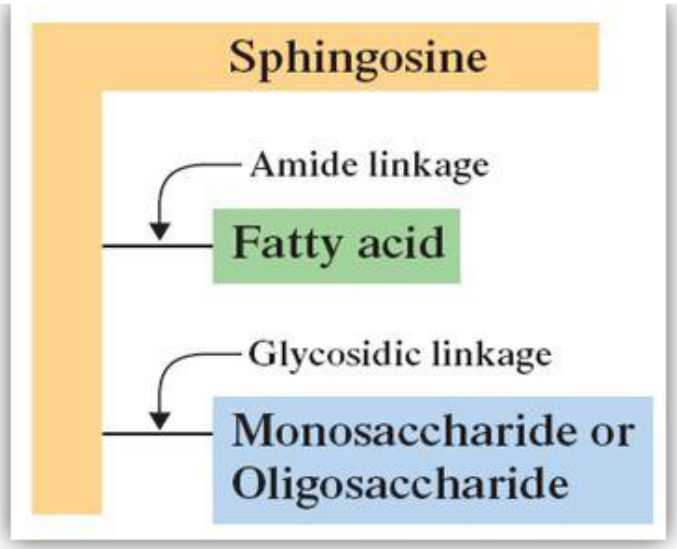
Name of sphingolipid	Name of X	Formula of X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	



Sphingoglycolipids

Or Glycolipids

Cerebrosides
(Monosaccharide)



Gangliosides
(Oligosaccharide)

are carbohydrate-containing ceramide derivatives
(in the outer face of plasma membranes)

Glycosphingolipids at the cell surface are sites of recognition.

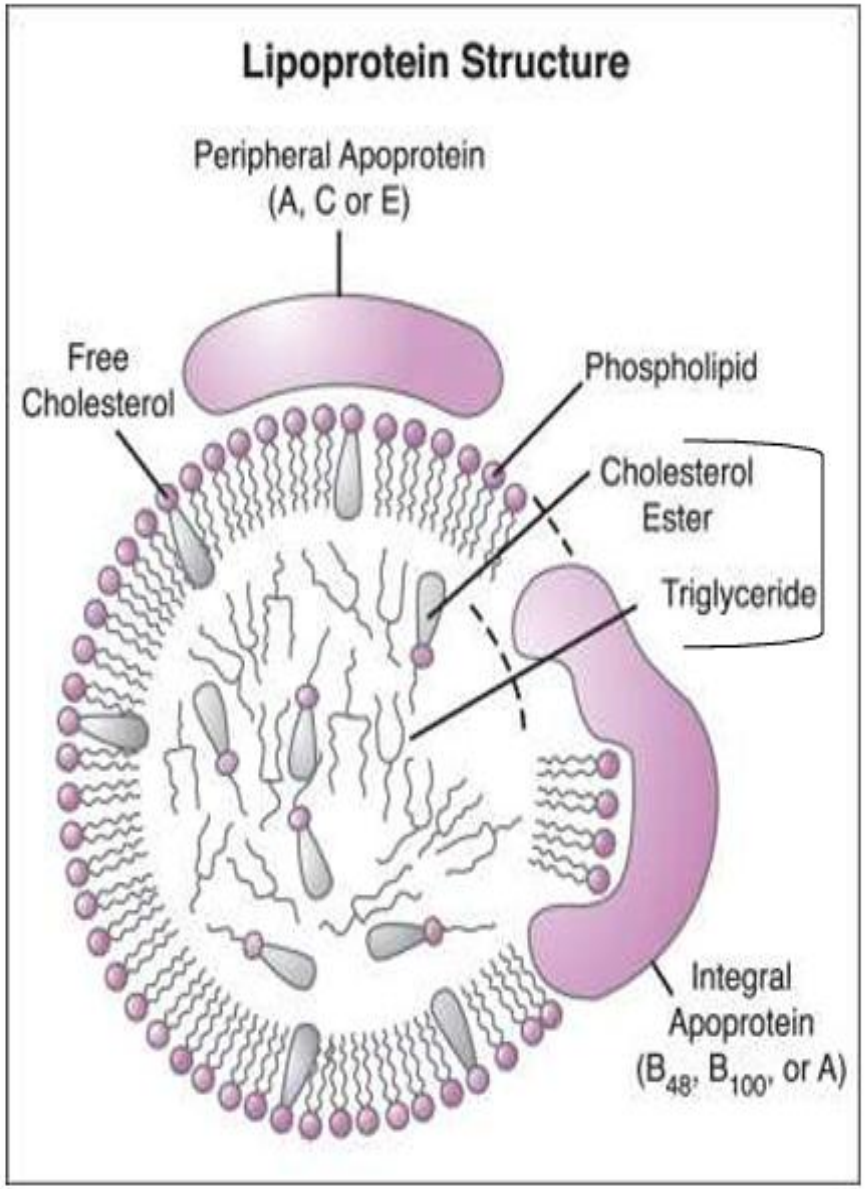
They found mainly in the myelin sheath and cell membrane of RBCs.

They act as cell membrane receptors for hormones and external stimuli.

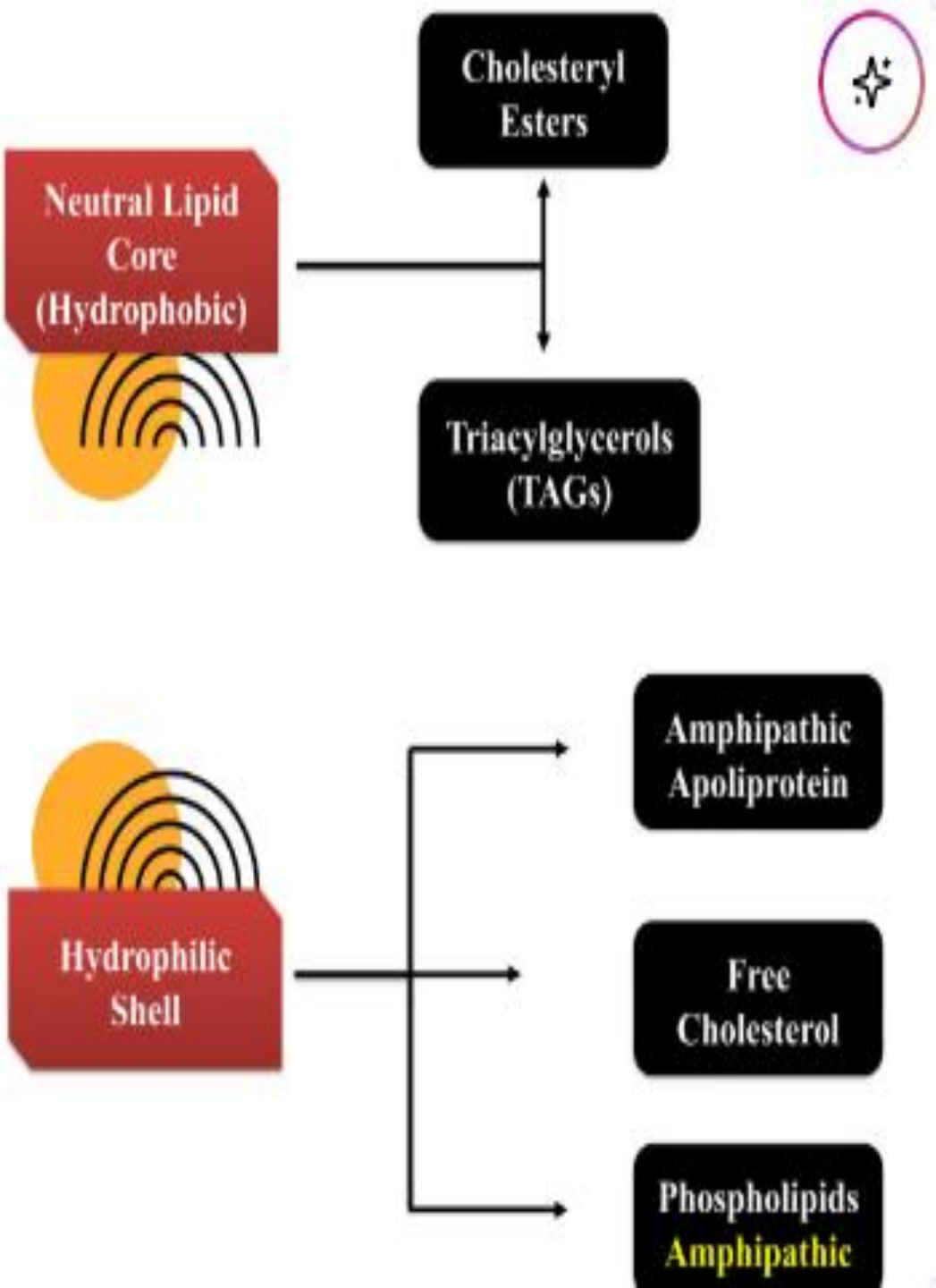
They provide recognition properties.

Lipoprotein***

- Lipoproteins are dynamic spherical macromolecular complexes of lipids and specific proteins that transport insoluble lipids through the blood between different organs and tissues.



Composition Of Lipoprotein



The Various Types of Lipoproteins and Their Composition

- There are various types of lipoproteins:
  - They differ in lipid and protein composition, therefore they differ in: Size, density and apoprotein content

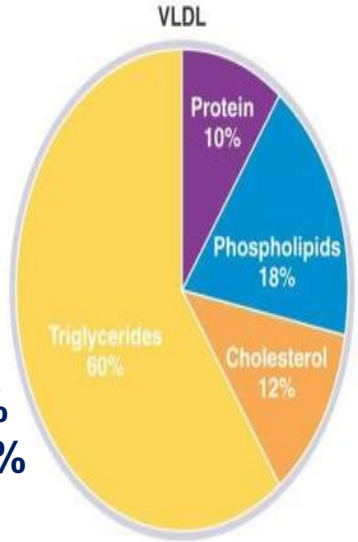
- They are:

Chylomicrons (CM)

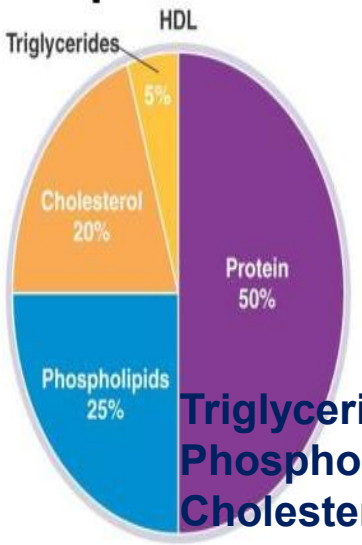
Very low density Lipoprotein VLDL)

Low density Lipoprotein (LDL)

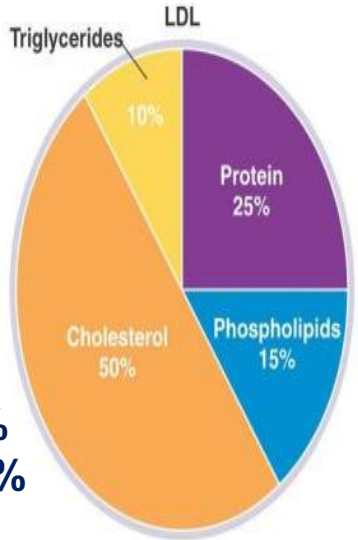
High density Lipoprotein (HDL)



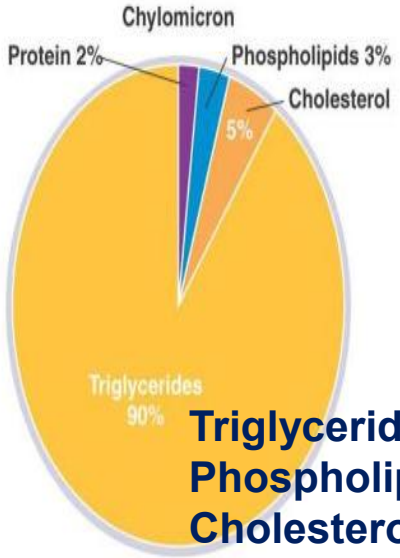
Triglycerides: 60%
Phospholipids:18%
Cholesterol: 12%



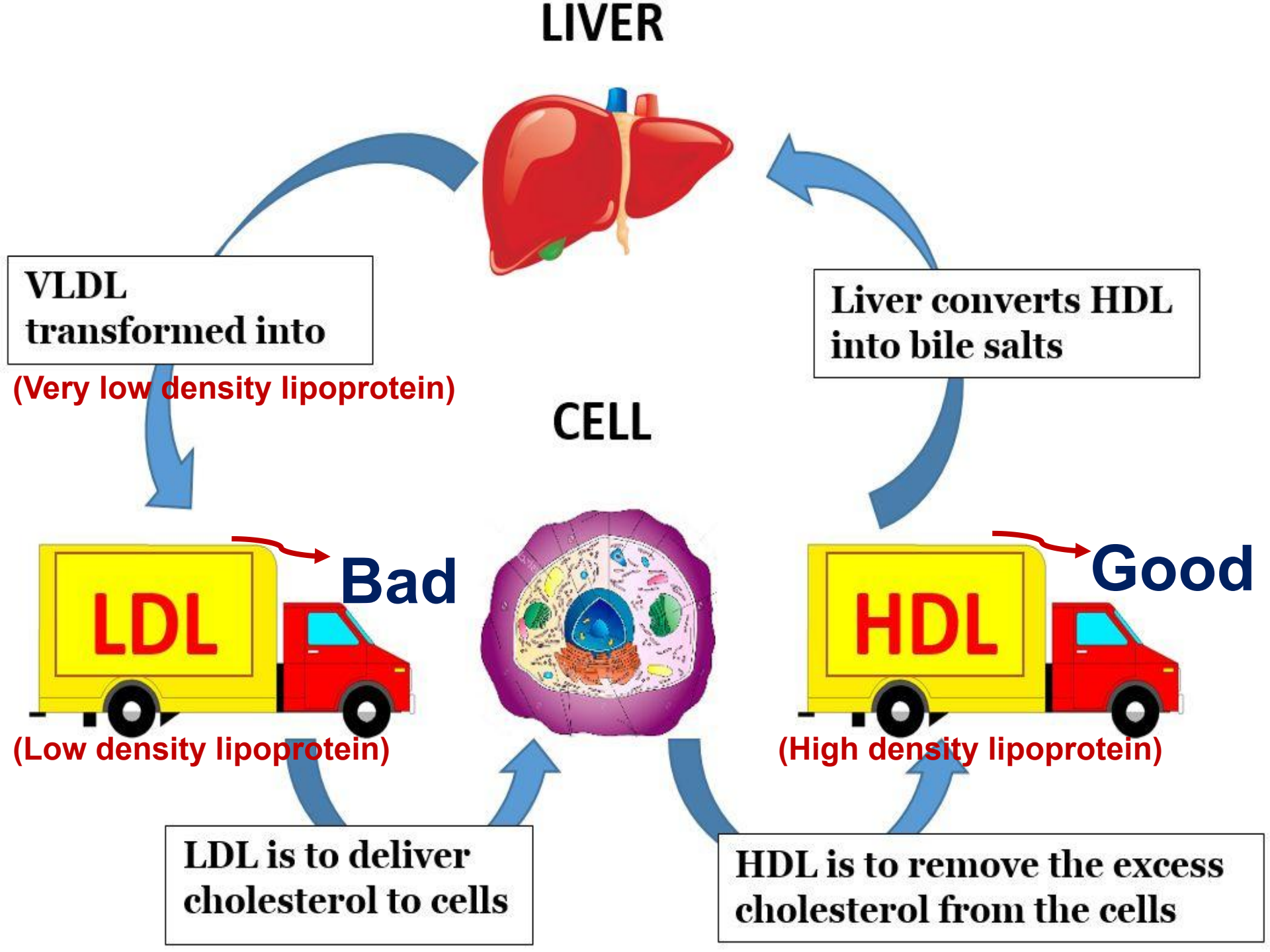
Triglycerides: 5%
Phospholipids:25%
Cholesterol: 20%



Triglycerides: 10%
Phospholipids:15%
Cholesterol: 50%



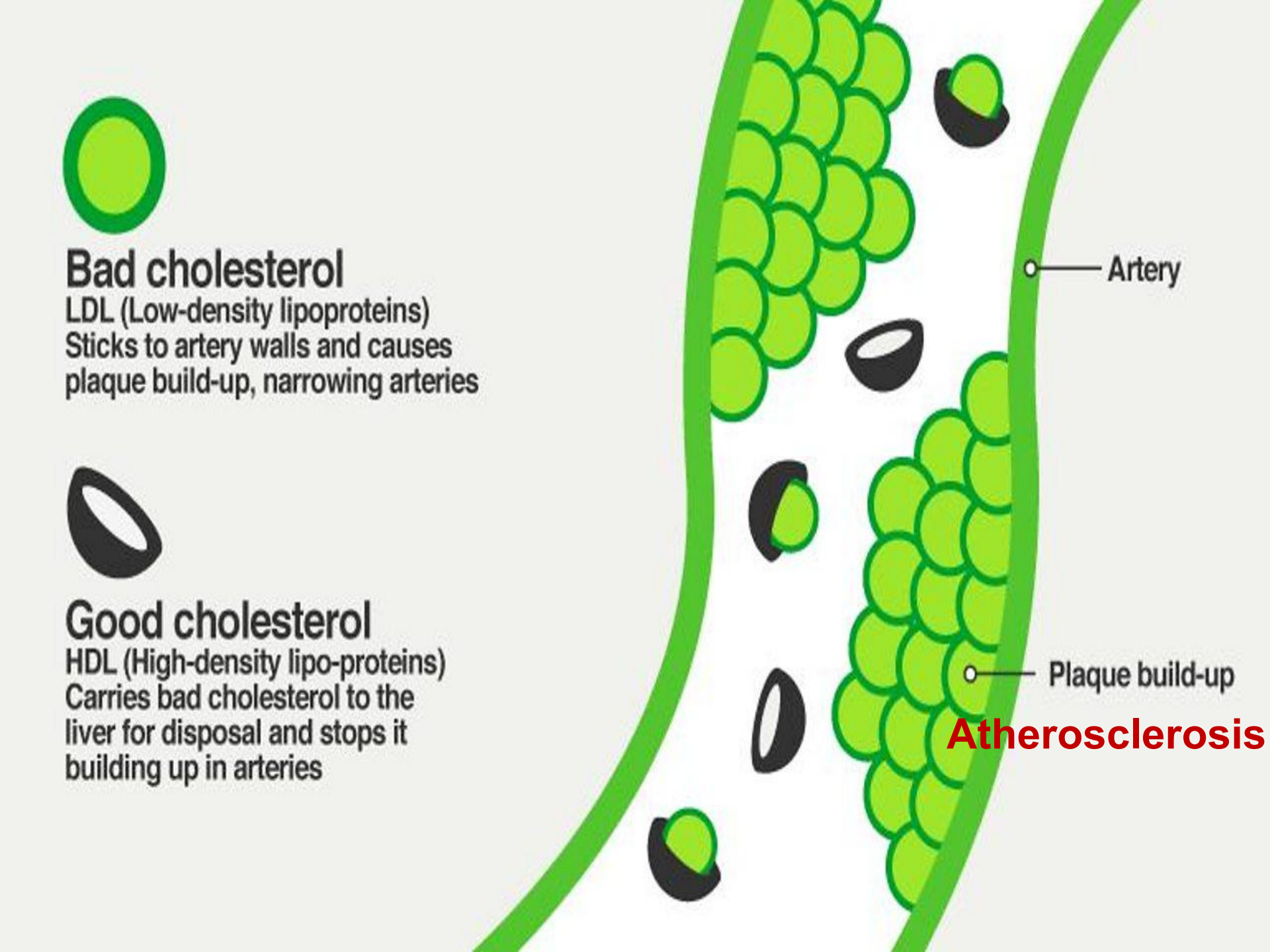
Triglycerides: 90%
Phospholipids:3%
Cholesterol: 5%


LIVER
VLDL transformed into
(Very low density lipoprotein)
Liver converts HDL into bile salts
CELL
LDL
Bad
(Low density lipoprotein)
HDL
Good
(High density lipoprotein)
LDL is to deliver cholesterol to cells
HDL is to remove the excess cholesterol from the cells

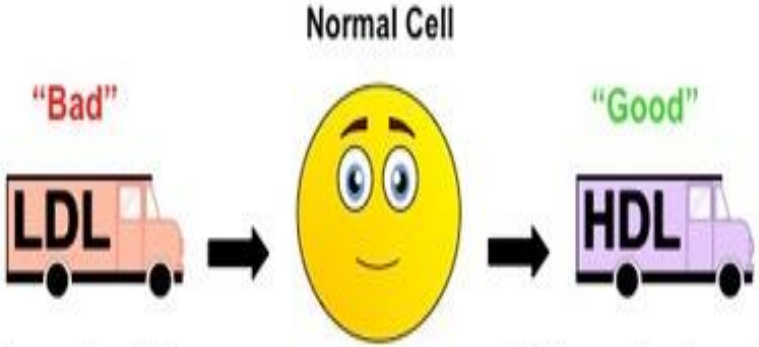
LDL is harmful but HDL is beneficial

- LDL is **Harmful** cause it deposits cholesterols in smooth muscle cells of artery & occurs atherosclerosis which **increase risk of cardiac disease (Myocardial infarction).**

- HDL is **beneficial** cause it transports free cholesterol from peripheral tissue to liver by reverse cholesterol transport for biliary excretion. So, **HDL keeps cholesterol concentration low in blood. It also has antiatherogenic property.**


Bad cholesterol
LDL (Low-density lipoproteins)
Sticks to artery walls and causes plaque build-up, narrowing arteries
Good cholesterol
HDL (High-density lipo-proteins)
Carries bad cholesterol to the liver for disposal and stops it building up in arteries
Artery
Plaque build-up
Atherosclerosis

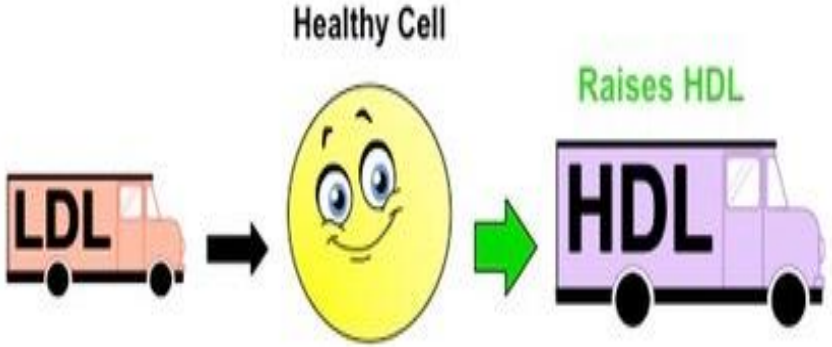
Normal Diet



LDL carries dietary fats into your cells

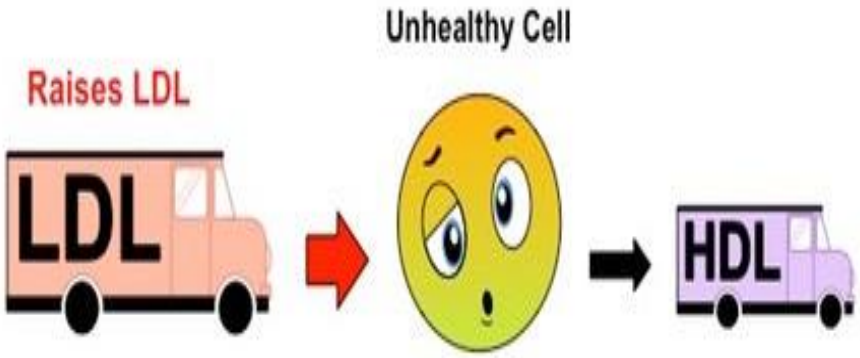
HDL carries impurities out of your cells

Diet Rich in *(CIS)* UNSATURATED FATS



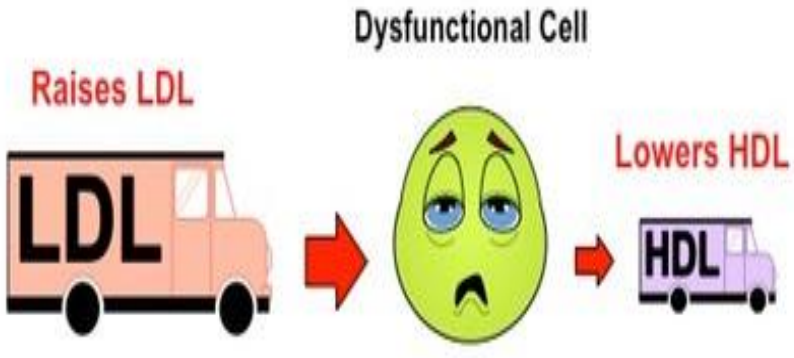
Diets rich in **unsaturated (*cis*) fats** *lower* cholesterol in the blood

Diet Rich in SATURATED FATS



Diets rich in **saturated fats** *raise* cholesterol in the blood

Diet Rich in *TRANS* FATS

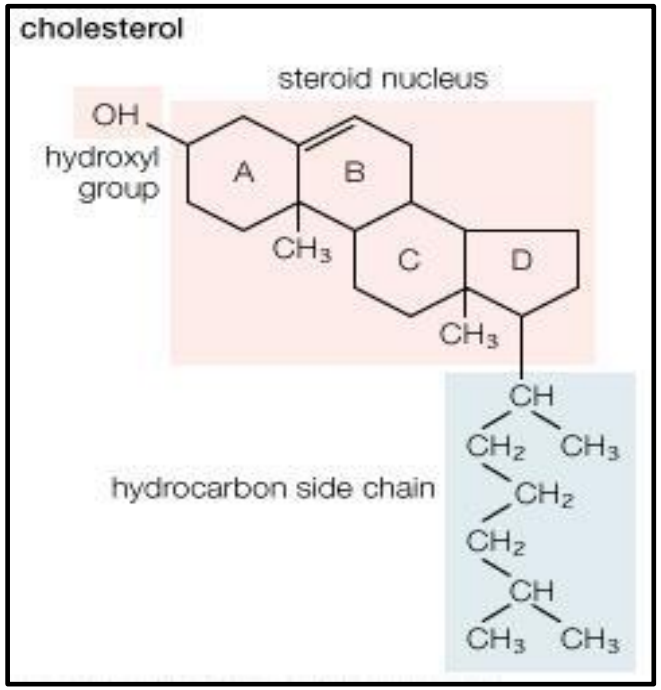


Diets rich in **trans fats** *significantly raise* cholesterol in the blood

Derived lipids—Steroids

- **Steroids** are lipids characterized by a carbon skeleton consisting of four fused rings
- **Cholesterol**, an important steroid, is a component in animal cell membranes
- Although cholesterol is essential in animals, high levels in the blood may contribute to cardiovascular disease

Cholesterol –a derived lipid



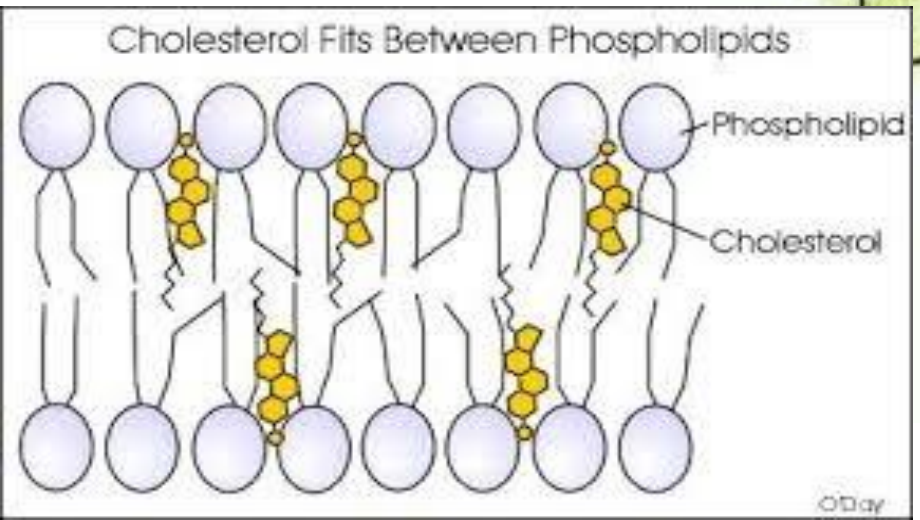
1.3.U3 Cholesterol is a component of animal cell membranes.

Cholesterol

Hydroxyl group makes the head polar and hydrophilic - attracted to the phosphate heads on the periphery of the membrane.

Carbon rings – it's not classed as a fat or an oil, **cholesterol is a steroid**

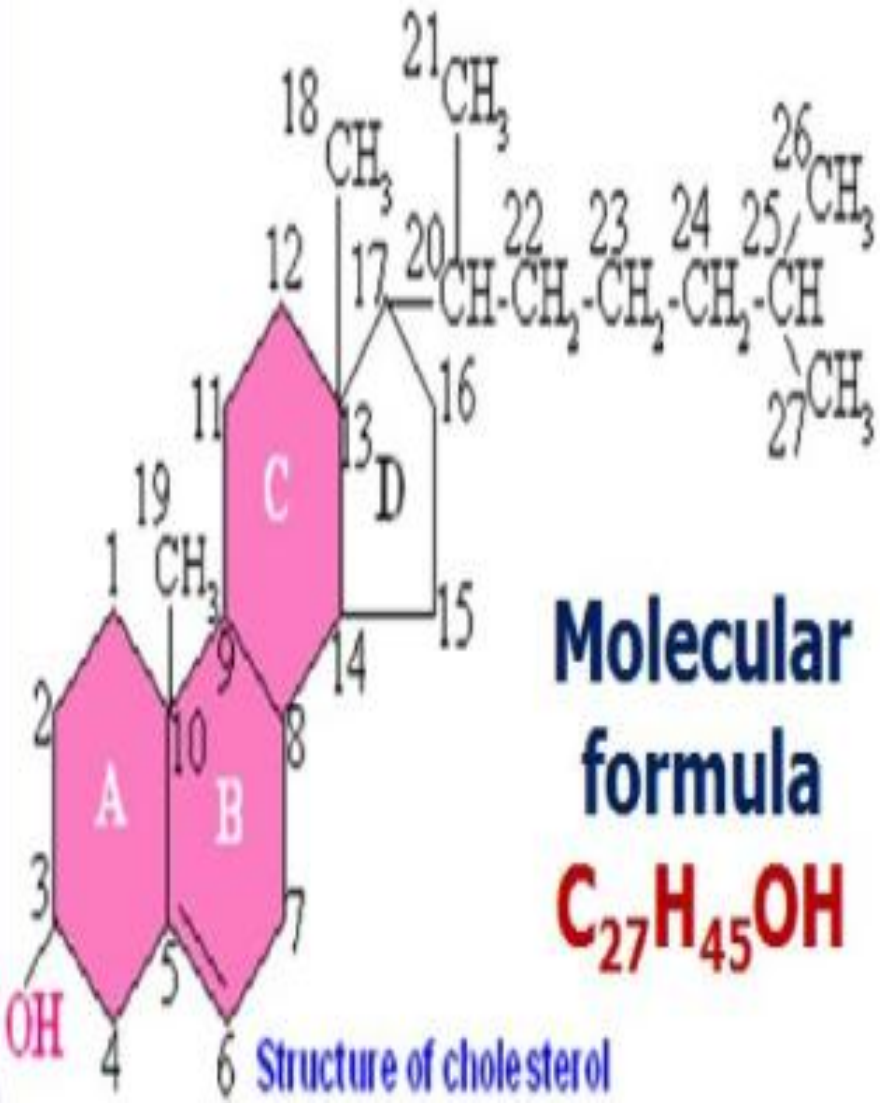
Non-polar (hydrophobic) tail –attracted to the hydrophobic tails of phospholipids in the centre of the membrane



http://www.uic.edu/classes/bios/bios100/lect03am/cholesterol.jpg

http://www.cholesterol-and-health.com/images/Cholesterol_Structure.jpg

Structure And Functions Of Cholesterol In The Body



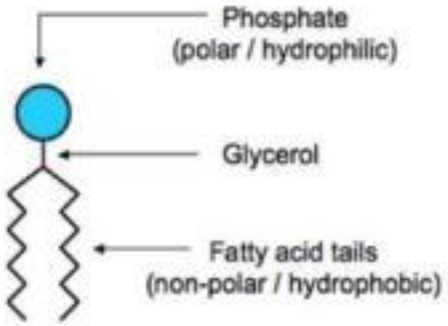
IMP

Functions

- Cholesterol is the structural component of the cell membrane.
- It's an essential ingredient in the structure of lipoproteins
- Fatty acid are transported to the liver as cholesteryl ester for oxidation
- It plays an important role in the synthesis of vitamin – D_3 which is responsible for proper bone calcification.
- Its s also helpful in healing process after a surgery.

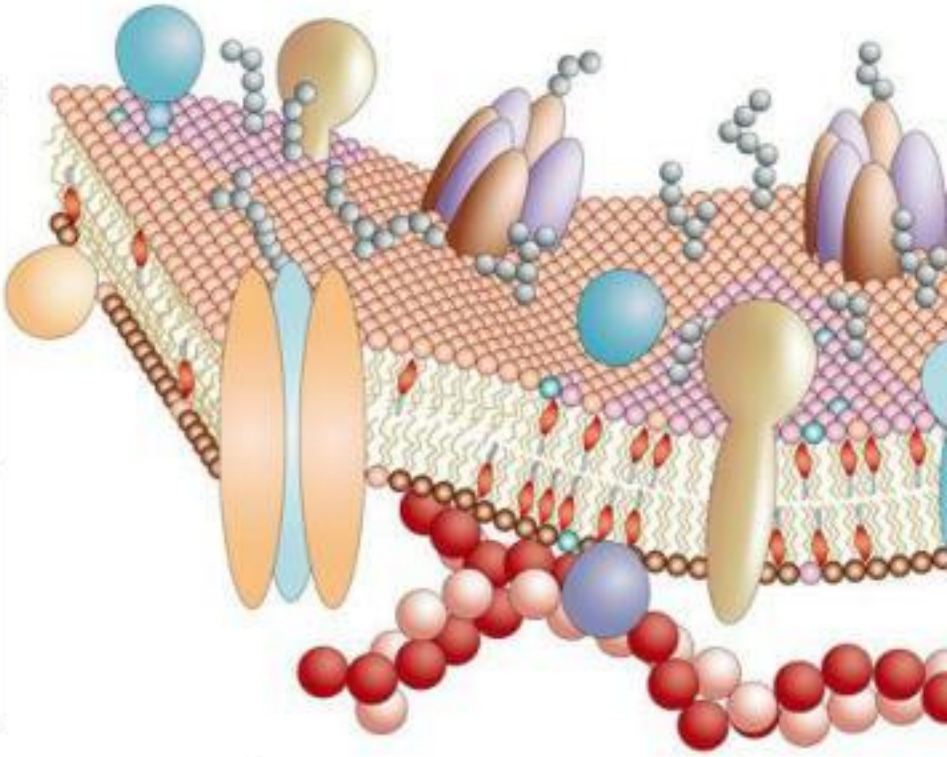
Cholesterol in mammalian membranes reduces membrane fluidity and permeability to some solutes.

Membrane fluidity



The hydrophobic hydrocarbon tails usually behave as a liquid. Hydrophilic phosphate heads act more like a solid.

Though it is difficult to determine whether the membrane is truly either a solid or liquid it can definitely be said to be fluid.



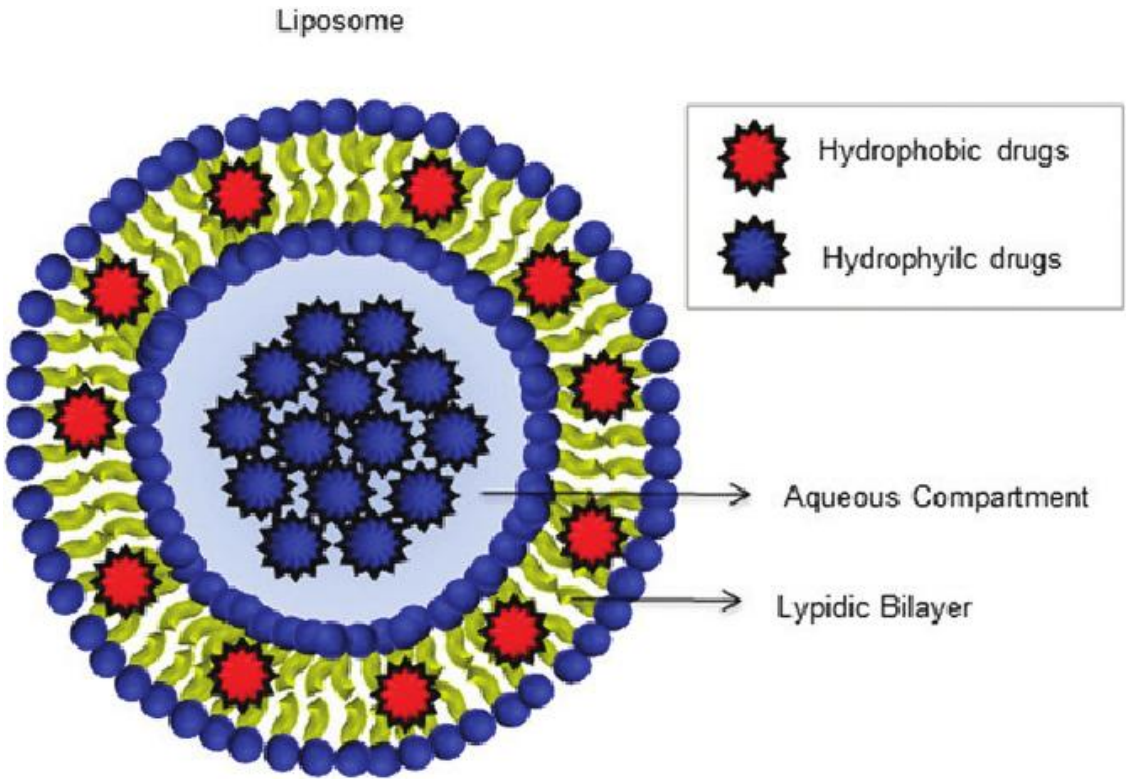
It is important to regulate the degree of fluidity:

- Membranes need to be fluid enough that the cell can move
- Membranes need to be fluid enough that the required substances can move across the membrane
- If too fluid however the membrane could not effectively restrict the movement of substances across itself

http://www.nature.com/scitable/content/ne0000/ne0000/ne0000/ne0000/14668965/U2.cp5.3_membrane_f2.jpg

Functions of lipids

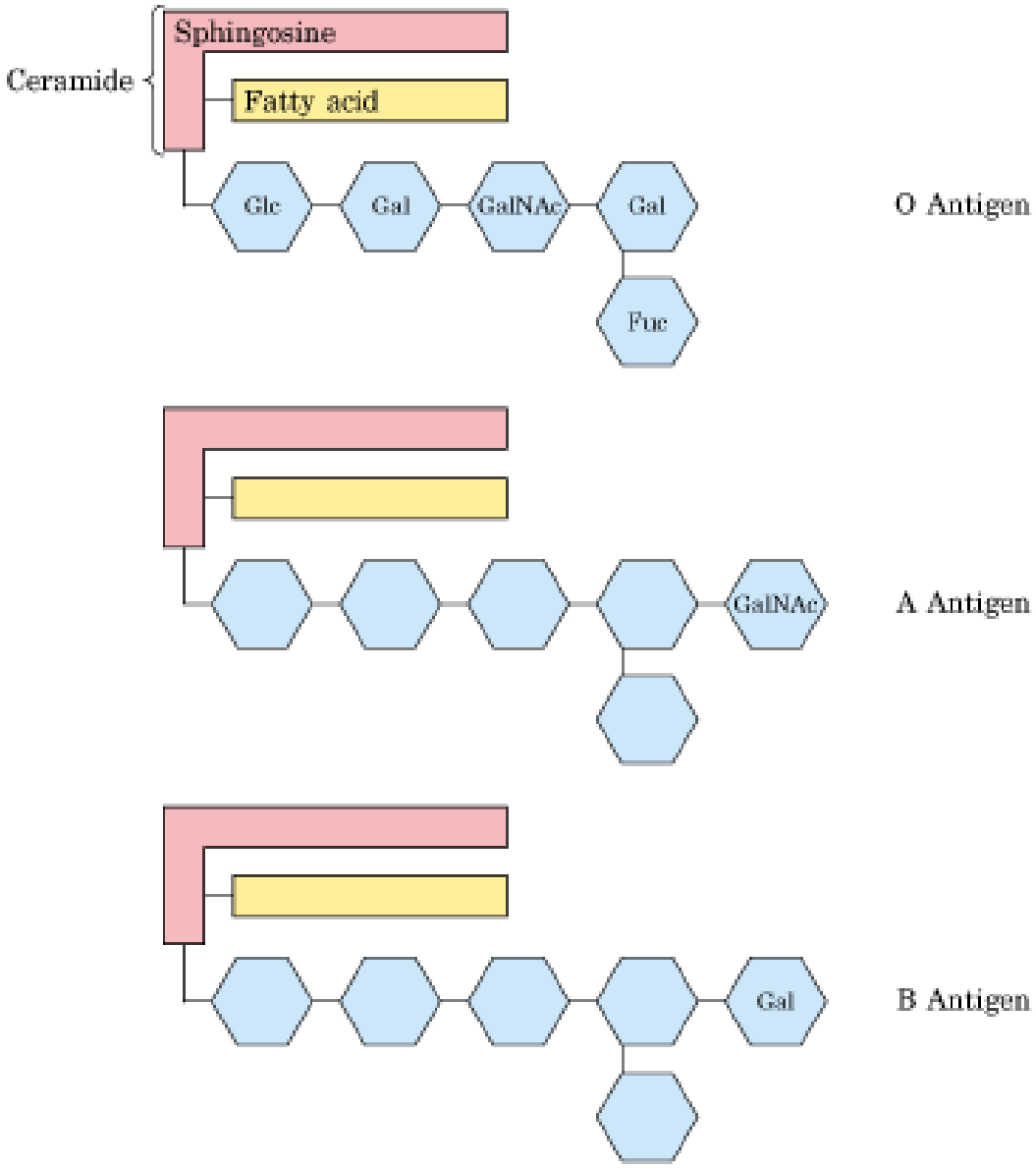
- Glycosphingolipids as blood group determinants
- Phospholipids are the major constituent of cell membrane.
- Lipids applications in drug delivery



Glycosphingolipids as blood group determinants

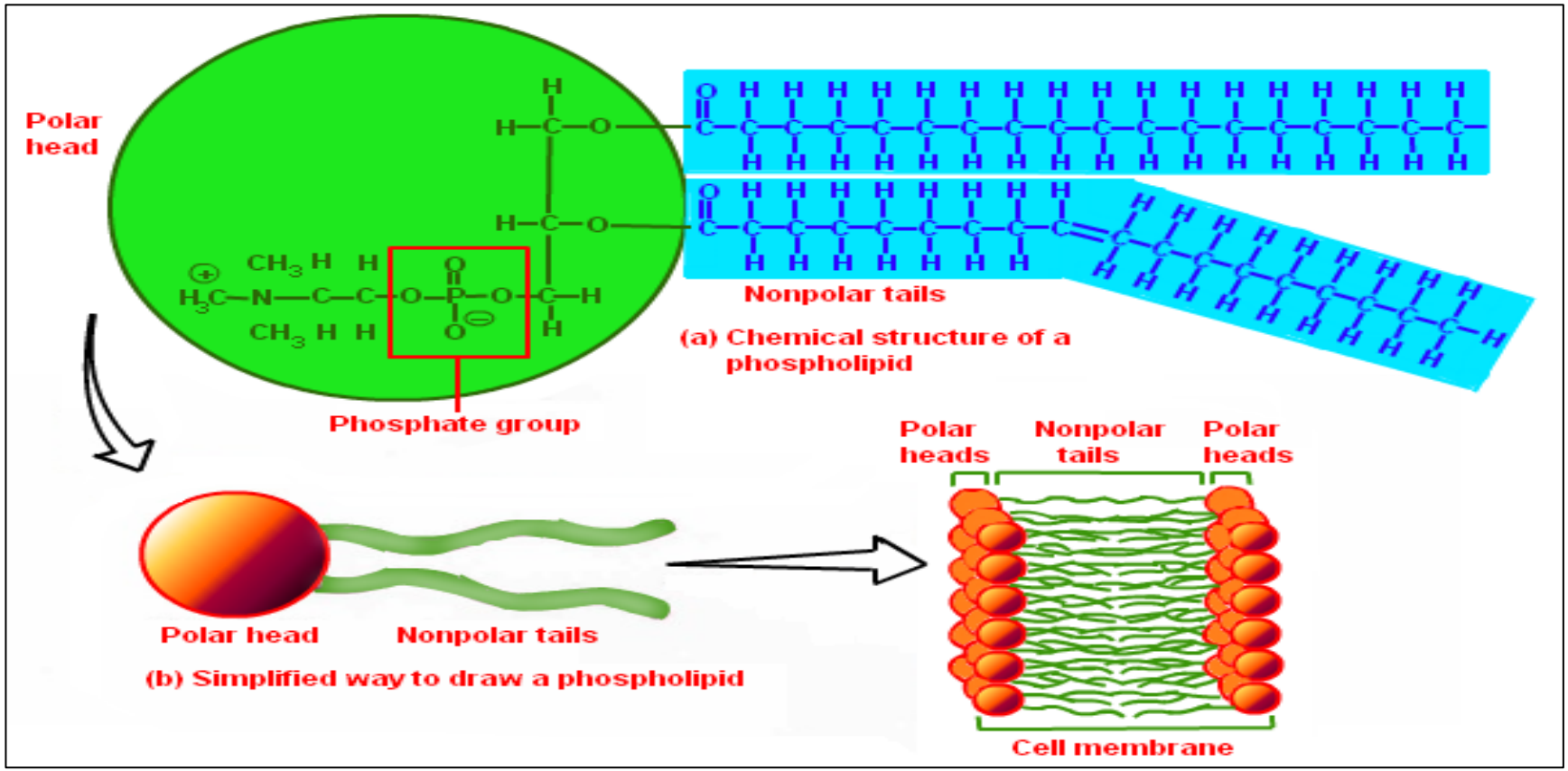
The human blood groups (O, A, B) are determined in part by the **oligosaccharide head groups** of these **glycosphingolipids.**

Glc:D-glucose
Gal:D-galactose
GalNAc:N-acetyl-D-galactosamine
Fuc:fucose

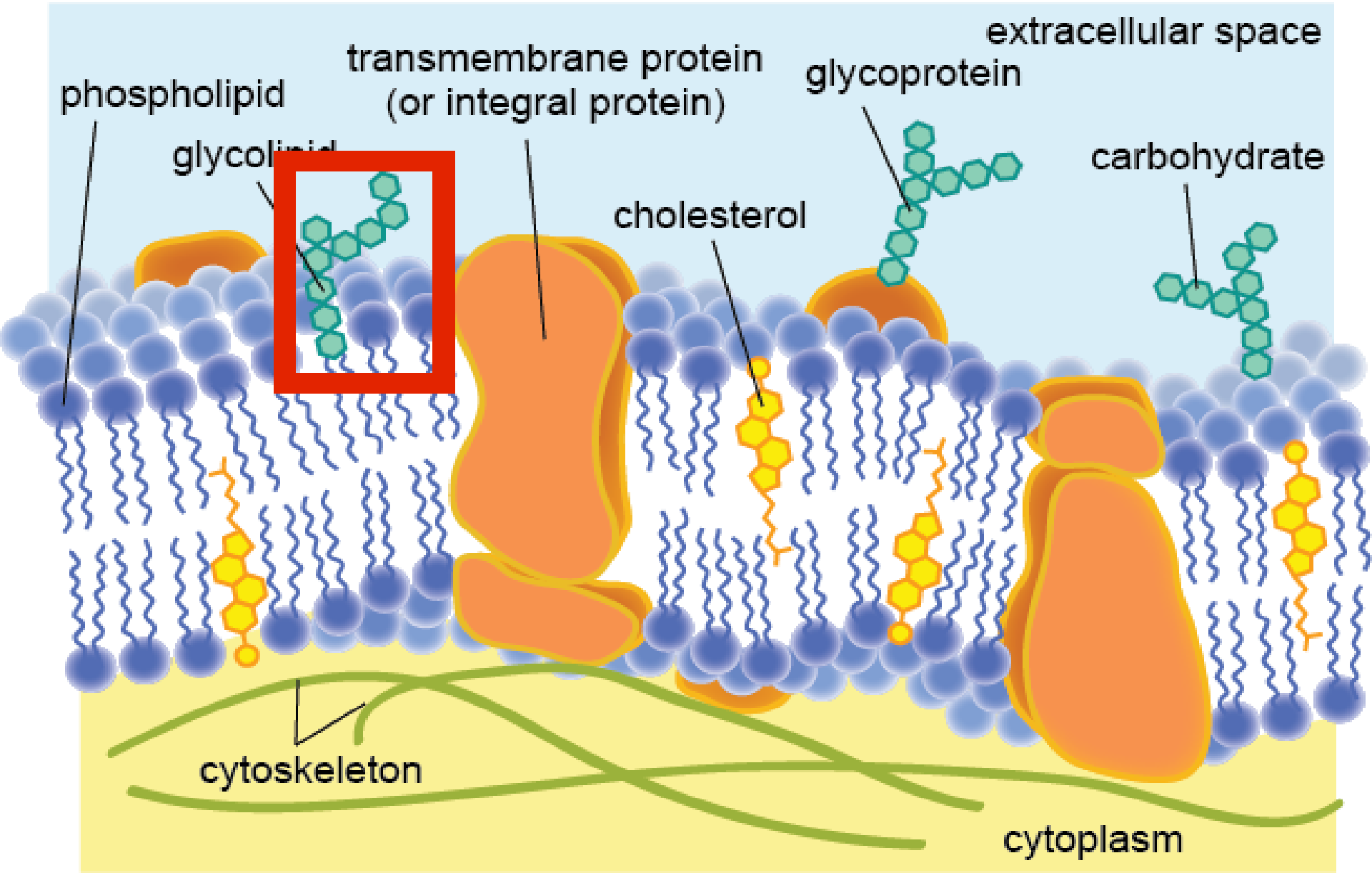


Lipids are the major constituent of cell membrane

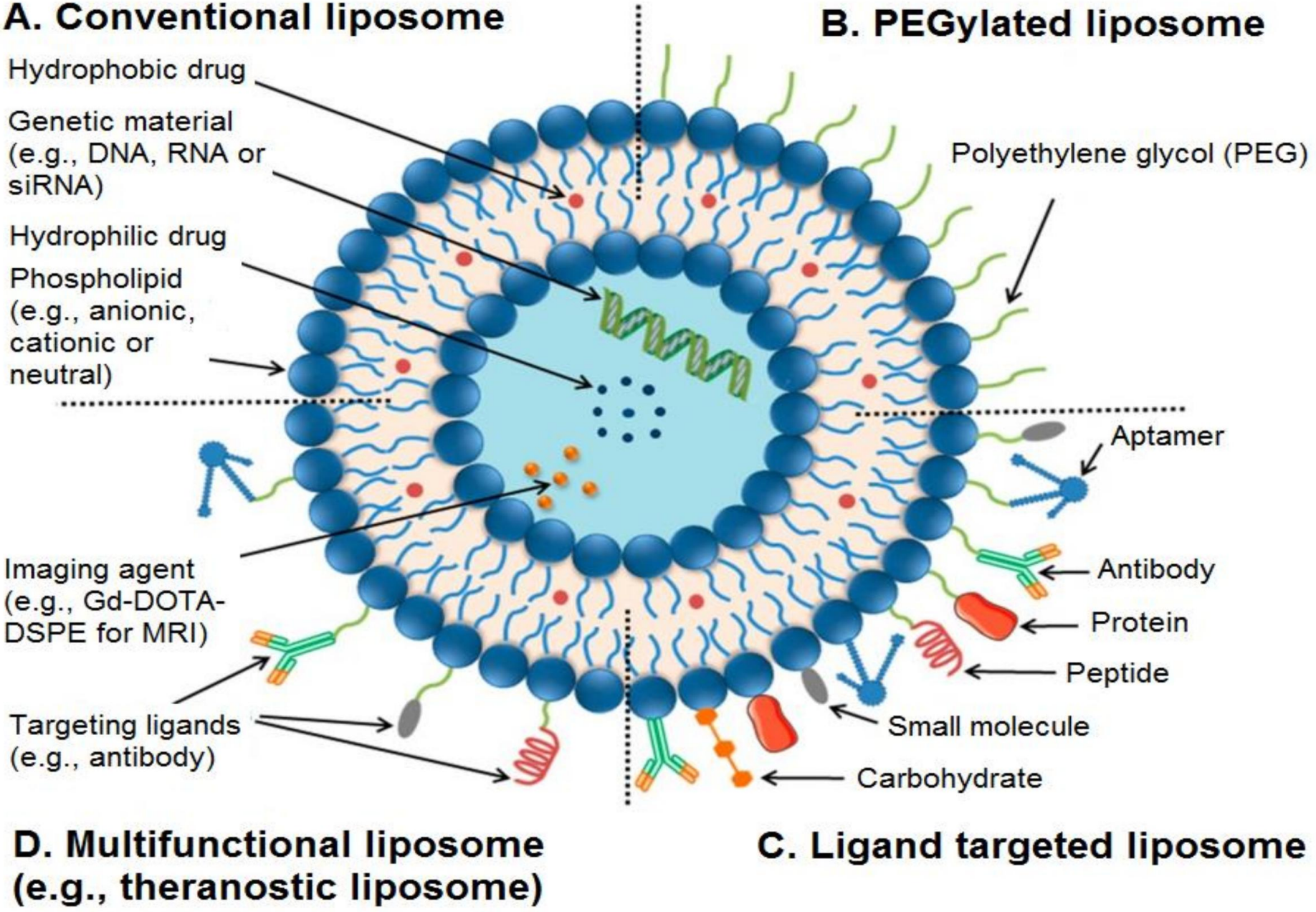
- **Plasma membrane is composed of phospholipids.**
- Phospholipids are arranged in two layers called **phospholipid bilayer**.



LIPID BILAYER STRUCTURE



Liposomes in drug/gene delivery



Sample questions

1. What is the solubility of lipids in water?

a) Soluble
b) Partially soluble
c) Insoluble
d) Partially insoluble

2. Fatty acids are amphipathic by nature.

a) True
b) False

3. Find the INCORRECT statement about the biological functions of lipids.

a) Storage form of metabolic fuel
b) Have a protective function in bacteria, plant, and insects
c) The structural component of membranes
d) Exhibit increased catalytic activity

4. The melting point of fatty acids depends upon chain length and _____

a) The shape of the fatty acids
b) The position of the double bond
c) Charge on the carbon
d) Degree of unsaturation

5. Which of the following is not a component of a phospholipid?

a) Phosphate
b) Alcohol
c) Glycerol
d) Protein

6. Which of the following phospholipid is considered as a major constituent of nervous tissue?

a) Glycerophospholipid
b) Plasmalogen
c) Inositol
d) Sphingomyelin

7. Identify the lowest density lipoprotein among the following?

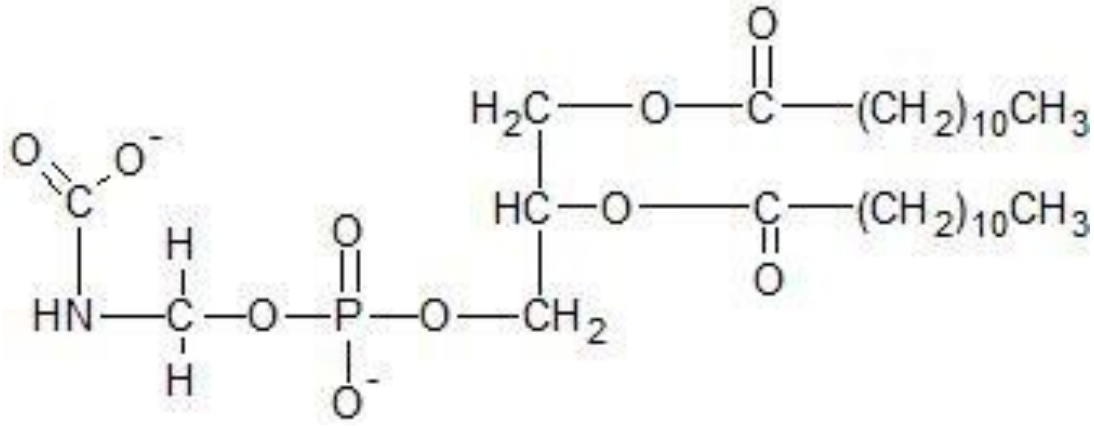
a) HDL
b) LDL
c) VLDL
d) Chylomicrons

8. The _____ the degree of unsaturation of the fatty acids of the bilayer, the _____ the temperature before the bilayer gels.

a) greater, lower
b) greater, more
c) lesser, higher
d) Lesser, lesser

Sample questions

(9). Examine the membrane lipid pictured below and answer the following questions.



I. Is this lipid classified as a phospholipid or a glycolipid? How can you tell?
II. Is this lipid considered a sphingolipid or a glycerophospholipid? How can you tell?
III. What fatty acid chains are used in this lipid? Are they saturated or unsaturated?
IV. What functional group enables them to connect to the backbone?

(10). Arrange the following fatty acids in order from lowest melting point to highest: myristic acid, arachidonic acid, linolenic acid, stearic acid, oleic acid.

A. **Myristic acid: 14 carbon saturated acid**
B. **Arachidonic acid: 20 carbon polyunsaturated acid (4 double bonds)**
C. **Linolenic acid: 18 carbon polyunsaturated acid (3 double bonds)**
D. **Stearic acid: 18 carbon saturated acid**
E. **Oleic acid: 18 carbon monounsaturated acid (1 double bond)**

(11). If a sample of a lipid contains fatty acids that are 89% saturated, would you expect this lipid to be solid at room temperature or liquid? What if the fatty acids were only 13% saturated?

(12) How can we differentiate between a glycerophospholipid and a sphingophospholipid?

(13) Steroid derivatives like cholesterol are also part of the lipid family. Name three useful by-products that cholesterol can be converted into within the body.